SYNTH4BENCH

Exploring Somatic Variant Callers' Behavior: A Synthetic Genomics Feature Space Approach

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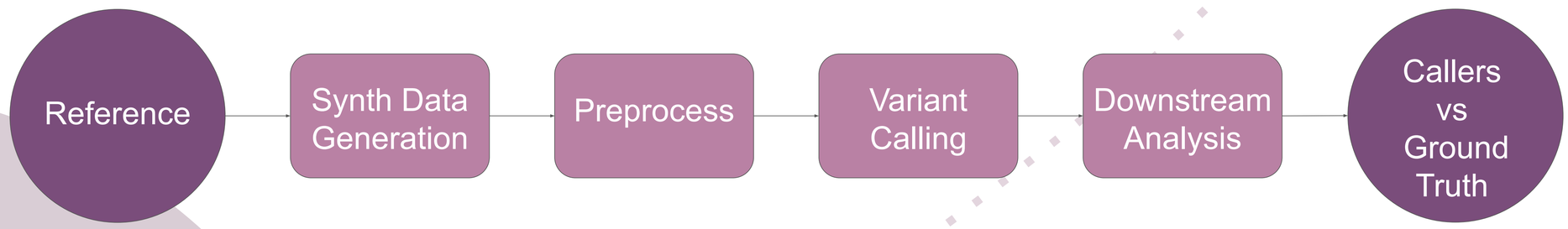
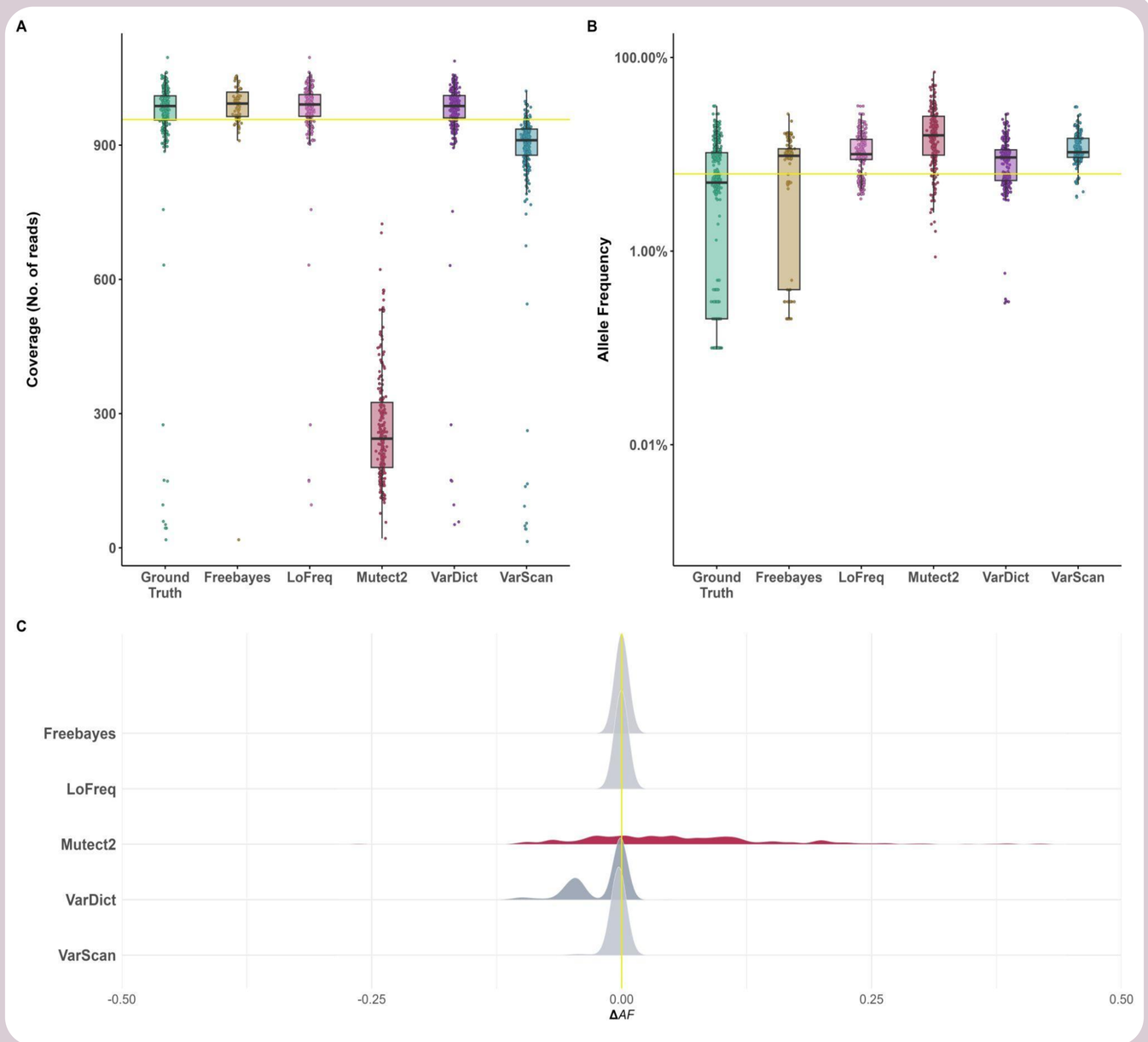


Background:
Identifying cancer-related genomic alterations with algorithms is vital, but evaluating their accuracy is challenging due to the lack of high-quality datasets. To tackle this issue, we implemented synth4bench, which calls the **NEATv3.3** simulator to generate synthetic genomics data, implements all somatic variant calling algorithms and then benchmarks them.

Methods:
We conducted a comprehensive evaluation of **Mutect2**, **Freebayes**, **VarDict**, **VarScan2**, and **LoFreq**, comparing their results with our produced synthetic ground truth. Synth4bench helps to elucidate the fundamental principles of each variant caller by exposing them to data across the genomics data feature space and inspecting their responses.

Results:
Using **synthetic datasets** as benchmarks is effective for evaluating **tumor-only somatic variant calling algorithms**. Synth4bench simplifies this process by providing a generated ground truth dataset, improving **precision** in cancer genomics, particularly for **low frequency variants**. This method enables **transparent and direct algorithm comparisons**, while exploring their **behavior**.



For 1000x depth and 150 read length dataset. **A** The **coverage boxplots** of ALT for the ground truth and all TP variants (those present in the ground truth that were also reported from a specific algorithm) for all callers. **B** The corresponding **AF boxplots** for the same TP variants of each scenario. **C** The **density plots** of the deviation $\Delta AF = AF_{Caller} - AF_{GT}$ for all TP variants.

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